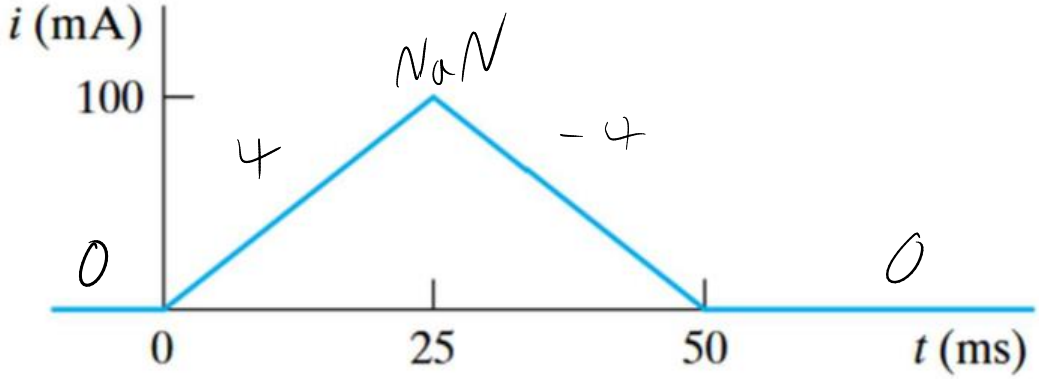
EE 2010
Fall 2026

Name Wyatt Evans

Game 29
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A current vs time waveform is shown below:



a) For a 68 mH inductor, find $V_L(t)$ for $t > 0$
b) For a 100 μF capacitor, find $V_C(t)$ for $t > 0$

$V_L = L\frac{di}{dt} = \frac{68}{1000}(\pm 4)$

(a) $V_L = 272mV, (0<t<25)$ $-272mV, (25<t<50)$ $0, (50<t<\infty)$

(b) $\frac{1}{100}\int_0^t i\,d\tau + V(t_0)$

$i(t) = 4t \quad \{0<t\leq 25\}$
$= 200-4t \quad \{25\leq t\leq 50\}$
$= 0 \quad \{50\leq t<\infty\}$

$V_c = \frac{t^2}{50}V, \quad 0\leq t\leq 25$

$\frac{-t^2+100t-1250}{50}V, \quad 25\leq t\leq 50$

$0\,V, \quad 50\leq t<\infty$